Recent advances on Ilex paraguariensis research: minireview.

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Ilex paraguariensis dried and minced leaves are made into a brewed tea, prepared in a sui generis manner by large populations in South America, having evolved from a tea drunk by the Guarani ethnic group to a beverage that has a social and almost ritualistic role in some South American modern societies. It is used both as a source of caffeine, in lieu or in parallel with tea and coffee, but also as a therapeutic agent for its alleged pharmacological properties. Although with some exceptions, research on biomedical properties of this herb has had a late start and strongly lags behind the impressive amount of literature on green tea and coffee. However, in the past 15 years, there was a several-fold increase in the literature studying Ilex paraguariensis properties showing effects such as antioxidant properties in chemical models and ex vivo lipoprotein studies, vaso-dilating and lipid reduction properties, antimutagenic effects, controversial association with oropharyngeal cancer, anti-glycation effects and weight reduction properties. Lately, promising results from human intervention studies have surfaced and the literature offers several developments on this area. The aim of this review is to provide a concise summary of the research published in the past three years, with an emphasis on translational studies, inflammation and lipid metabolism. Ilex paraguariensis reduces LDL-cholesterol levels in humans with Ilex paraguariensis dyslipoproteinemia and the effect is synergic with that of statins. Plasma antioxidant capacity as well as expression of antioxidant enzymes is positively modulated by intervention with Ilex paraguariensis in human cohorts. A review on the evidence implicating Ilex paraguariensis heavy consumption with some neoplasias show data that are inconclusive but indicate that contamination with alkylating agents during the drying process of the leaves should be avoided. On the other hand, several new studies confirm the antimutagenic effects of Ilex paraguariensis in different models, from DNA double breaks in cell culture models to mice studies. Novel interesting work has emerged showing significant effect on weight reduction both in mice and in rat models. Some mechanisms involved are inhibition of pancreatic lipase, activation of AMPK and uncoupling of electron transport. Intervention studies in animals have provided strong evidence of anti-inflammatory effects of Ilex paraguariensis, notably protecting cigarette-induced lung inflammation acting on macrophage migration and inactivating matrix-metalloproteinase. Research on the effects of Ilex paraguariensis in health and disease has confirmed its antioxidant, anti-inflammatory, antimutagenic and lipid-lowering activities. Although we are still waiting for the double-blind, randomized prospective clinical trial, the evidence seems to provide support for beneficial effects of mate drinking on chronic diseases with inflammatory component and lipid metabolism disorders.